

rithms, the key used for encryption cannot be used for decryption, it's safe to transmit the public key (the one which is to be used to encrypt the message). Any person who wants to send back a secure message can encrypt the message using your public key and you would be able to decrypt it using the private key!

DES

There is no way we can explain DES in detail within the limits of this book. Hence we would provide you with an introductory explanation of this algorithm. DES is actually an acronym for Data Encryption Standard.

DES is a block cipher which works by running an operation for 16 rounds on blocks of 64 bits. On each round, it takes the second half of the plaintext (32 bits) and transforms it into different text using a pre-designed algorithm. It then does an XOR operation of the result with the first half of the plaintext. The result of the XOR operation is processed by the algorithm (denoted as 'F' in the diagram) in the next step, while the second half is XORed with the result of the 'F' function in the next step. The method to get the plaintext is to reverse the procedure – to go from 'FP' towards 'IP' and reverse the algorithm.

RSA

The algorithm was designed by Ron Rivest, Adi Shamir and Leonard Adleman and it's their last names which make up the name of the algorithm. The RSA algorithm is given in following steps:

1. Two prime numbers p and q are taken.
2. A new number n is calculated by multiplying p and q

$$n = pq$$
3. Another number $\phi(n)$ is generated by using the formula:

$$\phi(n) = (p - 1)(q - 1)$$
4. Another integer e is to be found such that e lies between 1 and $\phi(n)$. Also, e and $\phi(n)$ must be coprimes. Mathematically

$$1 < e < \phi(n) \text{ where } \gcd(e, \phi(n)) = 1$$

(gcd = Greatest Common Divisor)
5. Another number d is to be found such that

$$de \bmod \phi(n) = 1$$

i.e. When the product of d and e is divided by $\phi(n)$, the remainder should be 1.